Alvin Ghindăoanu.

Industrial Engineering student

Bucharest, Romania alvinmarc17@yahoo.com

PROFILE

Final-year Industrial Engineering student at POLITEHNICA Bucharest, back from an Erasmus year at UPM Madrid. I design in CATIA, SolidWorks and AutoCAD, and I am looking for an internship or an entry-level engineering role.

EDUCATION

Bachelor's degree — Industrial Engineering 2023 – 2027
POLITEHNICA Bucharest — Faculty of Industrial Engineering and Robotics · Final year

Erasmus+ exchange year 2025 – 2026
Universidad Politécnica de Madrid (UPM) — ETSIDI
Project-based courses in mechatronics systems design, IoT with Raspberry Pi and photovoltaic-powered products.

Bachelor's degree — Business Administration (taught in foreign languages) 2023 – 2027
Bucharest University of Economic Studies (ASE) — FABIZ
Studied in parallel with engineering; paused for one year during the exchange.

Baccalaureate — Mathematics & Computer Science, intensive English 2023
I.L. Caragiale National College, Bucharest

PROJECTS

PipeBot — pipeline inspection drone 2025–26
UPM Madrid · Mechatronics Systems Design · Team of 4 · CATIA, Mechanical design, Design for 3D printing, Mechatronics
Design of a compact ducted-fan quadcopter for inspecting pipes and tanks; the frame doubles as a collision cage around the sensors and flight electronics.
My part: Co-designed the mechanical structure in CATIA and wrote the discussion, conclusions and cost analysis.
936 g take-off weight · 525 mm motor-to-motor span · ≈ €1,000 parts cost

Smart solar microgrid with wildfire early warning 2025–26
UPM Madrid · IoT and Raspberry Pi · Team of 2 · Raspberry Pi, Python, LoRa, Energy system sizing
Off-grid forest station that detects early signs of fire and raises the alarm over LoRa radio. Sized the solar, battery and supercapacitor system and checked it with a 24-hour simulation.
20 km LoRa link budget · 40+ days autonomy without sun

PV-powered key tracker 2026
UPM Madrid · Photovoltaic Powered Products · Team of 5 · Product design, Energy budget, Photovoltaics
Concept and energy budget for a key finder that never needs charging: indoor-light solar cell, Bluetooth and assisted GPS in a 40 mm disc.

Smart medicine box 2025
UPM Madrid · Practical IoT with Raspberry Pi · Team of 3 · Raspberry Pi, Python, Flask, MQTT
Working prototype that reminds a patient to take medication, confirms the box was opened and alerts a caregiver over MQTT.

SKILLS

CAD & CAM	SolidWorks, CATIA, AutoCAD, EdgeCAM
Software & prototyping	Python (basic), Raspberry Pi, Excel, PowerPoint, Word
Strengths	Discipline, Determination, Composure under pressure, Working in international teams, Adaptability

ADDITIONAL

Languages	Romanian (Native) · English (C1–C2) · German (A2)
Beyond engineering	Shotokan karate — World and European champion. More than 13 years of competition and still active. Certified national-level kumite referee. Volunteer — Chances for the Future (Șanse pentru viitor), 2022. Collected school and educational materials for disadvantaged children.
Courses	Circular Learning Space for Digital Skills, EIT RawMaterials (2022)
Licences	Driving B · Motorcycle A2
Interests	Snowboarding, Wakeboarding, Basketball